

# in-class template

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#1. Set up

## Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

## Data

```
# taxonomic information

taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```

# aquatic invertebrates

aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information

sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality

water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

```

## #2. Cleaning

### sites object

Goal: create a new object from `sites` that only includes NCOS sites

```

# creating new object from sites
ncos_sites <- sites |>
  # filtering to only include NCOS quarterly sampling sites
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

### clean taxon\_list

Goal: convert character strings in the existing `taxon_list` data frame into snakecase

```

# creating new object from taxon list
taxon_list_clean <- taxon_list |>
  # converted taxon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))

```

## clean water\_quality

Goal: create a data frame with median pH, median DO (mg/L), median salinity (ppt) for each date and site

- create a new object from `water_quality`
- clean column names
- only include surveys from quarterly NCOS sites
- create a new column called `date_site` to join with later data frames
- group by `date_site`
- calculate median pH, salinity, and dissolved oxygen
- filter out any outliers

```
# creating new data frame from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncoss_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, and DO
  summarize(
    med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup the data frame
  ungroup() |>
  # filtering out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

## clean aq\_ins

Goal: create a long format data frame in which each row contains the counted individuals in each taxon during each survey. The data frame should include taxonomic information for each organism and the water quality for each survey.

- create new object from `aq_ins`
- clean column names
- filter to only include quarterly NCOS sites
- select columns

- filter to only include “filtered beaker” sample
- convert the data frame to long format
- group by site, date, and taxon
- calculate average abundance per liter (sum counts divided by 7.5 liters)
- create new column called `date_site` to join with water quality
- join data frames
- join with taxon list

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
        ostracod,
        copepod,
        hemiptera_corixidae_boatman,
        cladocera,
        nematode,
        diptera_ceratopogonidae,
        annelida_oligochaete,
        diptera_chironomid,
        annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5 liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
```

```

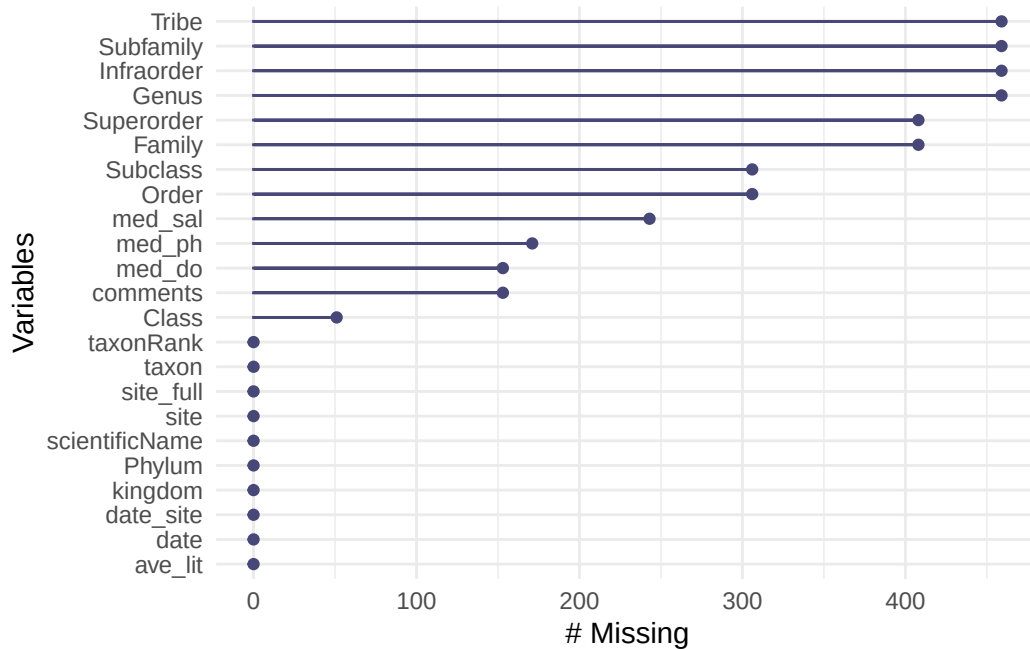
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm = TRUE),
       site_full = fct_rev(site_full))

```

### 3. Visualizing

#### Visualize missing values

```
gg_miss_var(aq_ins_clean)
```



Missing values for salinity, pH, and dissolved oxygen might influence later figures.

## Visualize salinity

- x-axis: `site`
- y-axis: `mean_sal`
- geometry to show salinity during survey: `geom_jitter`
- geometry to show mean salinity: `stat_summary`

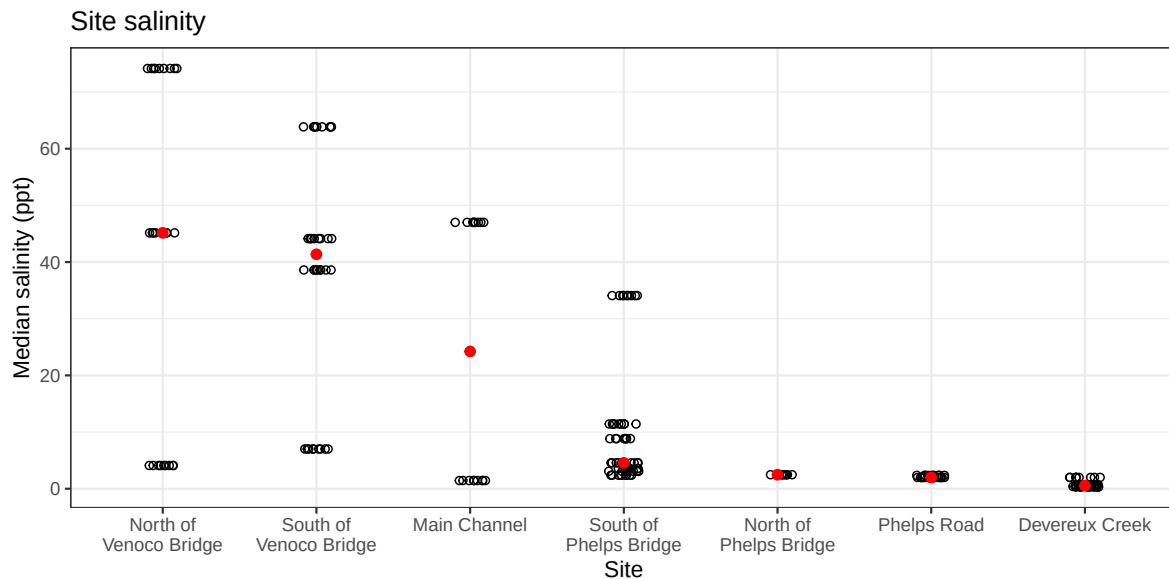
```
# creating a new object to store plot
salinity_plot <- ggplot(data = aq_ins_clean,
  # x-axis: full site name
  # y-axis: median salinity
  mapping = aes(x = site_full,
    y = med_sal)) +
  # points representing median salinity at each survey
  geom_jitter(height = 0,
    width = 0.1,
    shape = 21) +
  stat_summary(geom = "point",
    fun = median,
    color = "red",
```

```

      size = 2) +
# wrapping x-axis labels
scale_x_discrete(labels = label_wrap(width = 15)) +
labs(x = "Site",
     y = "Median salinity (ppt)",
     title = "Site salinity") +
theme_bw()

# displaying the object
salinity_plot

```



Sites seem to differ in salinity, North of Vencoco Bridge, East Channel, and Main Channel tend to be hypersaline, while Deveraux Creek, North of Phelps Bridge, Phelps Road are the least saline. South of Phelps Bridge tends to be more freshwater but is higher in salinity than the other, lower salinity sites.

## Visualize differences in copepod abundance

- x-axis: `site_full`
- y-axis: `ave_lit`
- geometry to represent observations: `geom_jitter()`
- geometry to represent medians: `stat_summary()`

## Initial cleaning

```
copepods <- aq_ins_clean |>
  filter(taxon == "copepod") |>
  mutate(log_ave_lit = log(ave_lit + 1))
```

## Visualizing:

```
# creating object to store plot
copepod_plot <- ggplot(data = copepods,
  # x-axis: full site name
  # y-axis: log transformed average count/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +

# jitter to represent each sample
geom_jitter(height = 0,
  width = 0.1,
  shape = 21) +

# points to represent median copepod abundance
stat_summary(geom = "point",
  fun = median,
  color = "blue",
  size = 2) +

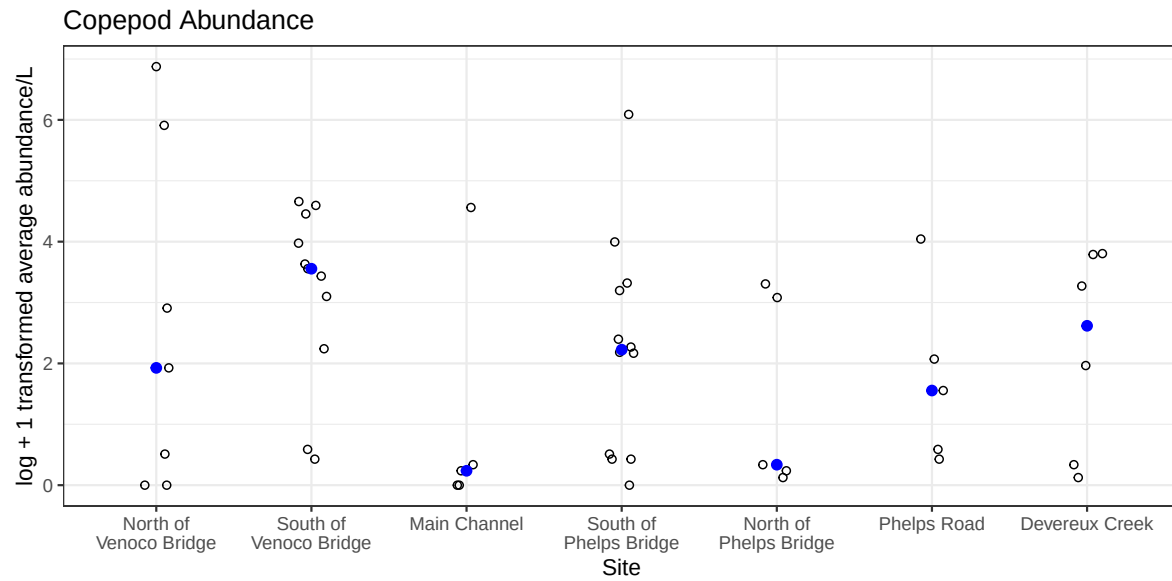
# wrapping x-axis labels
scale_x_discrete(labels = label_wrap(width = 15)) +

# relabelling axes
labs(x = "Site",
  y = "log + 1 transformed average abundance/L",
  title = "Copepod Abundance") +

# using different theme
theme_bw()

# displaying plot
copepod_plot
```





Copepod abundance differs across sites.

### Putting salinity and copepod abundance together

```
salinity_plot / copepod_plot
```

The map displays the North Fork of the Little River with sampling sites marked by black dots (1997) and red dots (2000). The sites are labeled as follows:

- North of Venoco Bridge
- South of Venoco Bridge
- Main Channel
- South of Phelps Bridge
- North of Phelps Bridge
- Phelps Road
- Devereux Creek

A legend indicates that black dots represent 1997 data and red dots represent 2000 data.

Scatter plot showing log + 1 transformed average abundance/L for various sites. The y-axis ranges from 0 to 6. The x-axis lists seven sites: North of Venoco Bridge, South of Venoco Bridge, Main Channel, South of Phelps Bridge, North of Phelps Bridge, Phelps Road, and Devereux Creek. Data points are represented by open circles (unexcavated) and filled blue circles (excavated).

| Site                   | Excavated (log + 1 transformed average abundance/L) | Unexcavated (log + 1 transformed average abundance/L) |
|------------------------|-----------------------------------------------------|-------------------------------------------------------|
| North of Venoco Bridge | 1.9                                                 | 0.0, 0.5, 2.9, 5.9, 6.8                               |
| South of Venoco Bridge | 3.5, 3.6, 3.7                                       | 0.4, 0.5, 2.2, 3.1, 3.5, 3.6, 3.7, 4.0, 4.6, 4.7, 4.5 |
| Main Channel           | 0.2                                                 | 0.0, 0.2, 0.3, 4.6                                    |
| South of Phelps Bridge | 2.2, 2.3                                            | 0.0, 0.4, 0.5, 2.2, 2.3, 2.4, 3.2, 3.3, 4.0, 6.1      |
| North of Phelps Bridge | 0.3                                                 | 0.1, 0.2, 3.1, 3.3                                    |
| Phelps Road            | 1.5                                                 | 0.4, 0.5, 1.5, 2.1, 4.0                               |
| Devereux Creek         | 2.6                                                 | 0.1, 0.3, 2.0, 3.3, 3.8                               |